

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Data Analysis and Visualization	Course Code:	DL3001
	Program:	BS Data Sciences	Semester:	Fall 2023
	Section	BS DS 5A	Total Marks:	15
	Date:	19th September 2023	Weight	
	Exam Type:	Quiz 1	Page(s):	2

Q1: Below is a dataset with some missing values, your task is to substitute suitable values in missing places using Linear Regression. **[5 Marks]**

Dataset:

Note: Data points format $\rightarrow (X, Y)$

Sr. No.	X	Y
1	1	2
2	3	_____
3	5	6
4	_____	8
5	9	10
6	11	12
7	13	_____
8	15	16
9	_____	18
10	19	20

X	Y	X ²	Y ²	X.Y
1	2	1	4	2
5	6	25	36	30
9	10	81	100	90
11	12	121	144	132
15	16	225	256	240
19	20	361	400	380
60	66	814	940	874

Solution:

Here, n=6

Since, $y=mx+c$

$$m = \frac{6(874) - (60 \cdot 66)}{6(814) - (60^2)} = 1.06$$

$$c = \frac{1}{6} \cdot (66 - (1.06 \cdot 60)) = 0.4$$

$$\text{So, } y = 1.06x + 0.4$$

Now, For $x=3$, $y=?$

$$y = 1.06(3) + 0.4 = 3.58$$

For $y=8$, $x=?$

$$x = \frac{(8 - 0.4)}{1.06} = 7.16$$

For $x=13$, $y=?$

$$y = 1.06(13) + 0.4 = 14.18$$

For $y=18$, $x=?$

$$x = \frac{(18 - 0.4)}{1.06} = 16.60$$

Q2: For the given filter on input, what will be the output in the image given below in case of convolution? **[10 Marks]**

Image:

0	0	0	0
0	0	0	0
0	0	1	0
0	0	0	0

Filter:

1	2	3
4	5	6
7	8	9

Solution:

Output Image:

1	2	3
4	5	6
7	8	9

Q1: Determine the noise value from the following dataset, and find a suitable substitute value to remove the noise from data using Linear Regression.

[5 Marks]

Dataset:

Note: Data points format → (X, Y)

Sr. No.	X	Y
1	1	4
2	2	5
3	3	6
4	4	<u>40</u>
5	5	9

Solution:

Noise: 40 in Y

X	Y	X^2	Y^2	XY
1	4	1	16	4
2	5	4	25	10
3	6	9	36	18
5	9	25	81	45

11	24	39	158	77
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Here, $n=4$

Since, $y=mx+c$

$$m = \frac{4(77) - (11 \cdot 24)}{4(39) - (11^2)} = 1.25$$

$$c = \frac{1}{4} \cdot (24 - (2.28 \cdot 11)) = 2.56$$

$$\text{So, } y = 1.25x + 2.56$$

Now, For $x=4$, $y=?$

$$y = 1.25(4) + 2.56 = 7.56$$

Q2: For the given filter on input, what will be the output in the image given below in case of convolution? **[10 Marks]**

Image:

0	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0

Filter:

9	8	7
6	5	4
3	2	1

Solution:

Output Image:

5	4
2	1